

such exercise is a finessing of the old popular saying that *even a broken clock is right twice a day*. We will take it a bit further to show that statistics is a knife that cuts on both sides. Let us use the Monte Carlo generator introduced earlier and construct a population of 10,000 fictional investment managers (the generator is not terribly necessary since we can use a coin, or even do plain algebra, but it is considerably more illustrative – and fun). Assume that they each have a perfectly fair game; each one has a 50% probability of making \$10,000 at the end of the year, and a 50% probability of losing \$10,000. Let us introduce an additional restriction; once a manager has a single bad year, he is thrown out of the sample, good-bye and have a nice life. Thus we will operate like the legendary speculator George Soros who was said to tell his managers gathered in a room: “half of you guys will be out by next year” (with an Eastern European accent). Like Soros, we have extremely high standards; we are looking only for managers with an unblemished record. We have no patience for low performers.

The Monte Carlo generator will toss a coin; *heads* and the manager will make \$10,000 over the year, *tails* and he will lose \$10,000. We run it for the first year. At the end of the year, we expect 5,000 managers to be up \$10,000 each, and 5,000 to be down \$10,000. Now we run the game a second year. Again, we can expect 2,500 managers to be up two years in a row; another year, 1,250; a fourth one, 625; a fifth, 313. We have now, simply in a fair game, 313 managers who made money for five years in a row. Out of pure luck.

NOBODY HAS TO BE COMPETENT

Let's push the argument further to make it more interesting. We create a cohort that is composed exclusively of incompetent managers. We will define an incompetent manager as someone who has a negative *expected return*, the equivalent of the odds being stacked against him. We instruct the Monte Carlo generator now to draw from an urn. The urn has 100 balls, 45 black and 55 red. By drawing with replacement, the ratio of red to black balls will remain the same. If we draw a black ball, the manager will earn \$10,000. If we draw a red ball, he will lose \$10,000. The manager is thus expected to earn \$10,000 with 45% probability,

and lose \$10,000 with 55%. On average, the manager will lose \$1,000 each round – but only *on average*.

At the end of the first year, we still expect to have 4,500 managers turning a profit (45% of them), the second, 45% of that number, 2,025. The third, 911; the fourth, 410; the fifth, 184. Let us give the surviving managers names and dress them in business suits. True, they represent less than 2% of the original cohort. But they will get attention. Nobody will mention the other 98%. What can we conclude?

The first counterintuitive point is that a population entirely composed of bad managers will produce a small amount of great track records. As a matter of fact, assuming the manager shows up unsolicited at your door, it will be practically impossible to figure out whether he is good or bad. The results would not markedly change, even if the population were composed entirely of managers who are expected in the long run to lose money. Why? Because owing to volatility, some of them will make money. We can see here that volatility actually helps bad investment decisions.

The second counterintuitive point is that the *expectation of the maximum* of track records, with which we are concerned, depends more on the size of the initial sample, than on the individual odds per manager. In other words the number of managers with great track records in a given market depends far more on the number of people who started in the investment business (in place of going to dental school), rather than on their ability to produce profits. It also depends on the volatility. Why do I use the notion of expectation of the maximum? Because I am not concerned at all with the average track record. I will get to see only the *best* of the managers, not all of the managers. This means that we will see more “excellent managers” in 2002 than in 1998, provided the cohort of beginners was greater in 1997 than it was in 1993 – I can safely say that it was.

ERGODICITY

To get more technical, I have to say that people believe that they can figure out the properties of the distribution from the sample they are witnessing. When it comes to matters that depend on the maximum, it is